A note on signature compression for SQIsign via Weil pairing recovery

Dustin Ray[1] and Paulo L. Barreto[2]

[1] Anchorage Digital, Security Engineering, USA

[2] University of Washington Tacoma, Tacoma, USA

Abstract. SQIsign round 3 transmits the full change-of-basis matrix of the response. This note states and proves, from the specification's own definitions, the Weil pairing relation that recovers one dropped entry of that matrix. It gives the pivot rule and a bit-packed layout, and says which pairing works and why. Sizes on the NIST known-answer vectors: 200 to 174 bytes at level I, 306 to 266 at level III and 406 to 352 at level V, 13%; the recovery adds 18% to a level I verification.

1 Purpose

Dropping one entry of a torsion-basis matrix and recovering it from the Weil pairing was introduced by Basso, Maino and Pope for FESTA [BMP23]. It is used in QFESTA [NO24] and in SQIsignHD [DLRW24]. The SIDH key-compression literature recovers scalars from pairings in the same way [CJL+17, ZSP+18, PB21]. The round-3 specification of SQIsign [AAA+26] transmits all four entries of $\mathbf{M}_{\mathrm{chl}}$ (Chapter 6). The formula for its two-dimensional response [BDF+24] had not been written down. We wrote it down for the maintainers on 8 July 2026, during round 2, as issue 19 of the reference repository [Ray26b]. Luca De Feo replied for the team that "we may introduce some compression in round 3, depending on performance". On 7 October 2026 he added that "we will give a closer look at compression for the next round". This note proves the formula from the specification's definitions and records what follows from it. That is a pivot rule, a bit-packed layout, the choice of pairing, and the sizes and costs measured with our Rust implementation [Ray26c] on the NIST known-answer vectors.

2 Notation

Curves, isogenies, the deterministic torsion bases and the Tate pairing are those of Chapters 3 and 4 of the round-3 specification [AAA+26], which writes f for the 2-adic valuation of $p+1$; we write F. The Weil pairing on an elliptic curve, which the round-3 specification uses only on abelian surfaces (Definition 4.5.3), is that of Section 2.2.5.1 of the v2.0 specification [AAA+25] and of [Sil86, III.8]. Per level: $\lambda = 128, 192, 256$; $c = 3, 27, 17$; $F = 324, 500, 664$; $e_{\mathrm{rsp}} = 196, 302, 400$.

E-mail: dustin@anchorage.com (Dustin Ray), pbarreto@uw.edu (Paulo L. Barreto)

Symbol	Meaning
p, F, c	$p = c \cdot 2^F - 1$ with c odd, $F = v_2(p+1)$; Section 5.2.
λ	The security parameter and the bit length of the challenge; Table 5.
e_{rsp}	$\lceil \log_2(p)/2 \rceil + \lambda/4 + 1$, the response exponent; Table 5.
n_{bt}, r_{rsp}	Zero in round 3. In v2.0, the backtracking count and $v_2(\deg \varphi_{\mathrm{rsp}})$, both in the signature ([AAA+25], Section 4.4.3); kept so that Theorem 1 covers both rounds.
N	$e_{\mathrm{rsp}} - n_{\mathrm{bt}} + 2$, the bit width of a matrix entry; Algorithm 3.24, line 3. Round 3: $e_{\mathrm{rsp}} + 2$.
g	$N - 1 = e_{\mathrm{rsp}} - n_{\mathrm{bt}} + 1$, the number of bits of $\det \mathbf{M}$ the pairings determine (Theorem 1). Round 3: $e_{\mathrm{rsp}} + 1$.
(P_0, Q_0)	The fixed basis of $E_0[2^F]$; Section 3.2.
$E_{\mathrm{pk}}, E_{\mathrm{com}}, E_{\mathrm{chl}}, E_{\mathrm{aux}}$	Public-key, commitment, challenge and auxiliary curves.
(A, B)	TorsionBasisFromHint$(E_{\mathrm{chl}}, e_{\mathrm{rsp}} + 2, \mathsf{hint}_{\mathrm{chl}})$, the deterministic basis of $E_{\mathrm{chl}}[2^N]$; Algorithm 3.24, line 10.
(C, D)	The same on E_{aux} (line 9), of order 2^N; in v2.0 of order $2^{N - r_{\mathrm{rsp}}}$.
$\mathbf{M}$	$\mathbf{M}_{\mathrm{chl}}$, entries $M_{00}, M_{01}, M_{10}, M_{11}$ in wire order, in $[0, 2^N)$ and normalised (lines 3 and 4).
P, Q	$(A, B) \cdot \mathbf{M} = ([M_{00}]A + [M_{10}]B,\ [M_{01}]A + [M_{11}]B)$; Algorithm 3.7 (ActOnBasis), called at line 11.
e_n, μ_n	The Weil pairing of order n; the n-th roots of unity in $\mathbb{F}_{p^2}^*$.
ω, η	$e_{2^N}(A, B)$ and $e_{2^N - r_{\mathrm{rsp}}}(C, D)$, both of exact order.

ActOnBasis takes the matrix columns as the coefficients of the new points, which is the reference implementation's convention; Section 1.4 of the v2.0 specification writes the same product with (A, B) as a column vector, which transposes the index names and changes nothing else.

3 The formula and its proof

What the matrix records. The signer (Algorithm 3.23 of [AAA+26], lines 15 to 19) produces a response $\varphi_{\mathrm{rsp}} : E_{\mathrm{com}} \to E_{\mathrm{chl}}$ of odd degree q (Algorithm 3.18; Section 9.5) and an auxiliary isogeny $\varphi_{\mathrm{aux}} : E_{\mathrm{com}} \to E_{\mathrm{aux}}$ of degree $2^{e_{\mathrm{rsp}}} - q$ (Algorithm 3.21, line 3). Algorithm 3.21 (lines 10 to 12) scales the commitment basis to $(P_{\mathrm{com}}, Q_{\mathrm{com}}) = [2^{F-N}]B_{\mathrm{com}}$, a basis of $E_{\mathrm{com}}[2^N]$, and computes the $(2^{e_{\mathrm{rsp}}}, 2^{e_{\mathrm{rsp}}})$-isogeny of the Kani square with sides φ_{rsp} and φ_{aux} (Algorithm 3.20), which returns the images of that basis on both codomains, $B_{\mathrm{chl}} = \varphi_{\mathrm{rsp}}(P_{\mathrm{com}}, Q_{\mathrm{com}})$ and $B_{\mathrm{aux}} = \varphi_{\mathrm{aux}}(P_{\mathrm{com}}, Q_{\mathrm{com}})$. Algorithm 3.9 (SetChangeOfBasisMatrix, called at line 18 of Algorithm 3.23) then computes the deterministic bases (C, D) of E_{aux} and (A, B) of E_{chl} with their hints, the matrix $\mathbf{M}_1$ with $B_{\mathrm{aux}} = (C, D) \cdot \mathbf{M}_1$, and the matrix $\mathbf{M}_{\mathrm{chl}}$ with $B_{\mathrm{chl}} \cdot \mathbf{M}_1^{-1} = (A, B) \cdot \mathbf{M}_{\mathrm{chl}}$. In v2.0 the response also has an even part of degree $2^{r_{\mathrm{rsp}}}$ and n_{bt} backtracking steps, and the auxiliary basis is scaled to order $2^{N - r_{\mathrm{rsp}}}$ ([AAA+25], Algorithms 4.2, 4.8 and 4.9); we carry r_{rsp} and n_{bt} through so that the statements below hold there too. In round 3 both are zero.

Lemma 1. *For a signature the signer produced there is a basis (R, S) of $E_{\mathrm{com}}[2^N]$ with $P = \varphi_{\mathrm{rsp}}(R)$, $Q = \varphi_{\mathrm{rsp}}(S)$, $C = \varphi_{\mathrm{aux}}([2^{r_{\mathrm{rsp}}}]R)$ and $D = \varphi_{\mathrm{aux}}([2^{r_{\mathrm{rsp}}}]S)$.*

Proof. Isogenies are linear, so with $(R, S) = (P_{\mathrm{com}}, Q_{\mathrm{com}}) \cdot \mathbf{M}_1^{-1}$, again a basis of $E_{\mathrm{com}}[2^N]$, Algorithm 3.9 gives $(P, Q) = (A, B) \cdot \mathbf{M}_{\mathrm{chl}} = B_{\mathrm{chl}} \cdot \mathbf{M}_1^{-1} = \varphi_{\mathrm{rsp}}(R, S)$ and $(C, D) = B_{\mathrm{aux}} \cdot \mathbf{M}_1^{-1} = \varphi_{\mathrm{aux}}(R, S)$. The x-only arithmetic determines each basis up to a common sign, which changes neither $\det \mathbf{M}$ nor the pairings below. In v2.0 the auxiliary basis of Algorithm 4.9, line 11, is the $[2^{r_{\mathrm{rsp}}}]$-multiple of $\varphi_{\mathrm{aux}}(R, S)$. □

Pairing facts. For n coprime to p, $T, U \in E[n]$ and an isogeny φ [Sil86, III.8.1], the following properties hold:

(W1) bilinearity: $e([\alpha]T, U) = e(T, [\alpha]U) = e(T, U)^{\alpha}$, $\forall \alpha \in \mathbb{Z}/n$;

(W2) alternation: $e_n(T, T) = 1$, hence $e_n(T, U) = e_n(U, T)^{-1}$;

(W3) $e_n(\varphi(T), \varphi(U)) = e_n(T, U)^{\deg \varphi}$;

(W4) for $T \in E[mn]$ and $U \in E[n]$, $e_{mn}(T, U) = e_n([m]T, U)$;

(W5) a basis of $E[n]$ pairs to a root of unity of order exactly n.

Theorem 1 (Weil pairing determinant recovery)**.** *For a signature the signer produced, $\omega^{\det \mathbf{M}} \cdot \eta$ is a primitive fourth root of unity. Consequently, with $\log_\omega$ the discrete logarithm in $\mu_{2^N} = \langle \omega \rangle$,*

$$\det \mathbf{M} \;\equiv\; \log_\omega\!\left(\eta^{-1}\right) + 2^{g-1} \pmod{2^g}, \qquad g = N - 1 = e_{\mathrm{rsp}} - n_{\mathrm{bt}} + 1.$$

The modulus is 2^g, the base is $\omega = e_{2^N}(A, B)$, and the argument is η^{-1}, not η.

Proof. Step 1. By (W1) and (W2),

$$\begin{aligned} e_{2^N}(P, Q) &= e_{2^N}([M_{00}]A + [M_{10}]B,\ [M_{01}]A + [M_{11}]B) \\ &= e_{2^N}(A, B)^{M_{00}M_{11}}\, e_{2^N}(B, A)^{M_{10}M_{01}} = \omega^{\det \mathbf{M}}. \end{aligned}$$

Step 2. Let $\rho = e_{2^N}(R, S)$ for the basis of Lemma 1; by (W5) it has order 2^N. By the lemma and (W3), $e_{2^N}(P, Q) = \rho^{\deg \varphi_{\mathrm{rsp}}} = \rho^{2^{r_{\mathrm{rsp}}} q}$. *Step 3.* By the lemma and (W3), $\eta = e_{2^{N-r_{\mathrm{rsp}}}}([2^{r_{\mathrm{rsp}}}]R, [2^{r_{\mathrm{rsp}}}]S)^{2^{e'}-q}$ with $e' = N - 2 - r_{\mathrm{rsp}}$. By (W4) with $m = 2^{r_{\mathrm{rsp}}}$, $n = 2^{N-r_{\mathrm{rsp}}}$, and then (W1),

$$e_{2^{N-r_{\mathrm{rsp}}}}([2^{r_{\mathrm{rsp}}}]R, [2^{r_{\mathrm{rsp}}}]S) = e_{2^N}(R, [2^{r_{\mathrm{rsp}}}]S) = \rho^{2^{r_{\mathrm{rsp}}}}, \qquad \text{so} \qquad \eta = \rho^{2^{r_{\mathrm{rsp}}}(2^{e'}-q)}.$$

Step 4. Multiplying, $\omega^{\det \mathbf{M}}\eta = \rho^{2^{r_{\mathrm{rsp}}}q + 2^{r_{\mathrm{rsp}}}(2^{e'}-q)} = \rho^{2^{e'+r_{\mathrm{rsp}}}} = \rho^{2^{N-2}} =: \zeta$; the sum of the degrees is the Kani condition of Algorithm 3.20 [Kan97]. Since ρ has order 2^N, ζ has order exactly 4. *Step 5.* $\eta^{-1} \in \mu_{2^N} = \langle \omega \rangle$, so $y = \log_\omega(\eta^{-1}) \in \mathbb{Z}/2^N\mathbb{Z}$ is defined, and $\zeta = \omega^{2^{N-2}u}$ for some $u \in \{1, 3\}$. Step 4 reads

$$\det \mathbf{M} \;\equiv\; y + 2^{N-2}u \pmod{2^N}, \qquad u \in \{1, 3\}. \tag{1}$$

Since u is odd, $2^{N-2}u \equiv 2^{N-2} \pmod{2^{N-1}}$, and reducing modulo $2^{N-1} = 2^g$ gives the formula. □

Corollary 1. *$v_2(\det \mathbf{M}) = r_{\mathrm{rsp}}$; in particular $\det \mathbf{M}$ is odd if and only if $r_{\mathrm{rsp}} = 0$, which is always the case in round 3.*

Proof. $\rho = \omega^t$ with t odd, both being generators of μ_{2^N}. Steps 1 and 2 give $\det \mathbf{M} \equiv t \cdot 2^{r_{\mathrm{rsp}}} q \pmod{2^N}$ with tq odd and $r_{\mathrm{rsp}} < N$. □

The one free bit. The specification carries the entries, hence $\det \mathbf{M}$, modulo 2^N, and Theorem 1 determines $\det \mathbf{M}$ modulo 2^{N-1}: exactly one bit of $\det \mathbf{M}$ is outside the pairings' reach, bit $N - 1$, which is the bit u of (1). The two missing bits of the naive count, N against the $N - 2$ bits of the $(2, 2)$-chain's length e', are the extra torsion the chain consumes (Section 4.5.4.1 of the specification), and Theorem 1 recovers one of them.

4 Recovering the dropped entry

Proposition 1. *For a signature the signer produced in round 3, M_{00} and M_{01} are not both even.*

Proof. $\det \mathbf{M} = M_{00}M_{11} - M_{01}M_{10}$ is odd by Corollary 1 with $r_{\mathrm{rsp}} = 0$. □

With $\det \mathbf{M} \bmod 2^g = y + 2^{g-1}$ from Theorem 1 and three entries, the fourth follows modulo 2^g. If M_{00} is odd the compressor drops M_{11} and the decompressor computes

$$M_{11} \equiv \left(\det \mathbf{M} + M_{01}M_{10}\right) \cdot M_{00}^{-1} \pmod{2^g}. \tag{2}$$

If M_{00} is even, M_{01} is odd by Proposition 1; the compressor then drops M_{10} and the decompressor computes

$$M_{10} \equiv \left(M_{00}M_{11} - \det \mathbf{M}\right) \cdot M_{01}^{-1} \pmod{2^g}. \tag{3}$$

The decompressor reads the path off the parity of the transmitted M_{00}; the inverse is a Newton iteration from $M_{00}^{-1} \equiv 1 \pmod 2$, doubling the precision each step. The top bit of the dropped entry, bit $N - 1 = e_{\mathrm{rsp}} + 1$, is outside the recovery; verification never reads it (§5), so the format sends nothing for it.

5 Format

The round-3 specification [AAA+26] has primes $3 \cdot 2^{324} - 1$, $27 \cdot 2^{500} - 1$, $17 \cdot 2^{664} - 1$ (Section 5.2, after [Wes26]), a response of odd degree with no even part and no backtracking count (Algorithm 3.18; Section 9.5), $e_{\mathrm{rsp}} = 196, 302, 400$ (Table 5), both bases of order $2^{e_{\mathrm{rsp}}+2}$ (Algorithm 3.24, lines 9 and 10), and the matrix normalised to the lexicographically smaller of $\mathbf{M}$ and $-\mathbf{M}$ (Algorithm 3.10, checked at line 4 of Algorithm 3.24; the sign freedom of [Ray26a]). The signature is $(\mathsf{chl}, E_{\mathrm{aux}}, \mathbf{M}_{\mathrm{chl}}, \mathsf{hint}_{\mathrm{aux}}, \mathsf{hint}_{\mathrm{chl}})$ in $2 \cdot \mathtt{FP} + 4\lceil (e_{\mathrm{rsp}} + 2)/8 \rceil + \lambda/8 + 2$ bytes (Chapter 6, Table 11), with $\mathtt{FP} = \lceil \log_2(p)/8 \rceil = 41, 64, 84$. Theorem 1 applies with $N = e_{\mathrm{rsp}} + 2$ and $g = e_{\mathrm{rsp}} + 1$.

The compressed signature. Table 1 is the compressed signature: one bit string, bit i of the string being bit $i \bmod 8$ of byte $\lfloor i/8 \rfloor$, each field the low bits of its little-endian integer. The two coordinates of A_{aux} take $\lceil \log_2 p \rceil$ bits each, the three transmitted matrix entries $e_{\mathrm{rsp}} + 1$ bits each, the challenge λ bits, the two hints 8 bits each, and the string is padded with zero bits to a byte boundary. The decoder rejects a wrong length, a coordinate at or above p, a non-zero padding bit, and M_{00} and M_{01} both even.

Table 1: The round-3 compressed signature: $2\lceil \log_2 p \rceil + 3(e_{\mathrm{rsp}} + 1) + \lambda + 16$ bits, zero-padded to 174 / 266 / 352 bytes.

Field	Bits (I / III / V)	Content
A_{aux}	2× 326 / 505 / 669	real then imaginary part of the Montgomery coefficient, each in $[0, p)$
M_{00}, M_{01}, M_{var}	3× 197 / 303 / 401	the low $e_{\mathrm{rsp}} + 1$ bits; $M_{\mathrm{var}} = M_{10}$ if M_{00} is odd, else M_{11}
chl	128 / 192 / 256	the challenge, as in the standard format
$\mathsf{hint}_{\mathrm{aux}}$, $\mathsf{hint}_{\mathrm{chl}}$	8 + 8	as in the standard format
padding	5 / 1 / 3	zero
total	1392 / 2128 / 2816	174 / 266 / 352 bytes

The dropped entry is recovered modulo $2^g = 2^{e_{\mathrm{rsp}}+1}$, so its bit e_{rsp} is computed. Bit e_{rsp} of the three transmitted entries is carried because the $(2, 2)$-chain consumes P and Q

themselves, and its acceptance depends on that bit on most signatures. Bit $e_{\mathrm{rsp}}+1$, the top bit, is not needed, for two reasons in the specification. Adding $2^{e_{\mathrm{rsp}}+1}$ to an entry adds a point of order 2 to P or Q; the chain of Algorithm 3.20 uses the two points only through their multiples, the kernel $\langle [4]P, [4]Q\rangle$ [AAA+26, §4.5.4.1 and Algorithm 4.132] and the 8-torsion points above it that the gluing consumes [AAA+26, §4.5.4.2], which a 2-torsion summand does not change. And the sign check of Algorithm 3.24, line 4, reads that bit only on the first entry that is neither 0 nor $2^{e_{\mathrm{rsp}}+1}$, where the signer's normalisation leaves it zero. The decompressor sets bit $e_{\mathrm{rsp}}+1$ to zero in every entry.

Sizes. Table 2 gives the standard and compressed sizes. Decompression is a deterministic public map, and the compressed verifier accepts exactly when the standard verifier accepts the reconstructed signature, so the security reduction for the standard format [ABF+25] applies unchanged. The reconstructed matrix may differ from the signer's in bit $e_{\mathrm{rsp}}+1$; this is immaterial since SQIsign does not target strong unforgeability [AAA+26, Section 8.4.2]. The two hint bytes could be dropped as well, since the decompressor builds both curves and can run TorsionBasisToHint on them (the recomputed hints equal the transmitted ones on 300 of 300 vectors, §8), for 172 / 264 / 350 bytes; we keep them, as the specification does, because they are what makes the verifier's bases cheap (Section 3.4.3).

Table 2: Signature sizes in bytes on the NIST known-answer vectors.

Level	Standard	Compressed	Saved
I	200	174	26 (13%)
III	306	266	40 (13%)
V	406	352	54 (13%)

6 Pairing selection

On these curves the reduced Tate pairing of Section 4.4.4 of the specification is an explicit power of the Weil pairing. Everything in this section follows from that.

Lemma 2. *Let $E/\mathbb{F}_{p^2}$ be supersingular with $\#E(\mathbb{F}_{p^2}) = (p+1)^2$, and write $p+1 = c\,2^F$ with c odd. For $1 \le m \le F$ and all $T, U \in E[2^m]$,*

$$t_{2^m}(T,U) \;=\; e_{2^m}(T,U)^{-c\,2^{F-m}}.$$

Proof. The p^2-power Frobenius π satisfies $\pi^2 + 2p\,\pi + p^2 = 0$, since the trace of $E/\mathbb{F}_{p^2}$ is $-2p$. Thus $(\pi+p)^2 = 0$, and as $\mathrm{End}(E)$ has no nonzero nilpotents, $\pi = [-p]$. By the Frobenius description of the reduced Tate pairing [Sch05], for $T \in E(\mathbb{F}_{p^2})[n]$, $U \in E(\mathbb{F}_{p^2})$ and any V with $[n]V = U$,

$$t_n(T,U) = e_n\big(T,\ \pi(V) - V\big).$$

Take $n = 2^m$. Then $\pi(V) - V = -(p+1)V = -c\,2^{F-m}\,[2^m]V = -c\,2^{F-m}\,U$, and the claim follows by bilinearity of e_{2^m}. □

Remark 1. Under the opposite normalization of either pairing the exponent changes sign; the proof of Proposition 2 uses only that $t_{2^m} = e_{2^m}^{\kappa\,2^{F-m}}$ for an odd κ that is the same on every curve $\mathbb{F}_{p^2}$-isogenous to E_0, all of which are maximal.

The Tate pairing is alternating here. By Lemma 2 and (W2), for every $T \in E[2^m]$

$$t_{2^m}(T,T) = e_{2^m}(T,T)^{-c\,2^{F-m}} = 1,$$

so the reduced Tate pairing is alternating on these curves, and Step 1 of the proof of Theorem 1 reads $\det \mathbf{M}$ off t_{2^m} as off e_{2^m}.

The transmitted basis is divisible. (A, B) has order 2^N and $2N > F$ at every level. For $N \le m \le F$, (W4) gives $e_{2^m}(A, B) = e_{2^N}([2^{m-N}]A, B) = \omega^{2^{m-N}}$, so by Lemma 2

$$t_{2^m}(A, B) = \omega^{-c\,2^{F-N}} \qquad \text{for every } m \ge N,$$

a value independent of the ladder's length, of order 2^{2N-F}: 2^{72}, 2^{108}, 2^{140} at levels I, III, V, against the $g = 197, 303, 401$ bits of $\det \mathbf{M}$ needed. The same holds for (C, D). On the 300 KAT vectors the two values have exactly that order at every m from N to F, and their discrete logarithm never determines $\det \mathbf{M}$ modulo 2^g, or modulo $2^{e_{\mathrm{rsp}}}$, at any m. No Tate ladder on the bases the verifier has recovers the entry; the Weil pairing, which does not divide by 2^{F-N}, is required on them.

One Tate ladder per curve on the full-order bases. What a ladder of length m loses to the divisibility it regains from the points of order 2^F that the hints also define.

Proposition 2. *Let (R, S) and (R', S') be the bases of $E_{\mathrm{chl}}[2^F]$ and $E_{\mathrm{aux}}[2^F]$ that* TorsionBasisToHint(E, F, N) *returns (Algorithm 3.11; Algorithm 4.94), so that $(A, B) = [2^{F-N}](R, S)$ and $(C, D) = [2^{F-N+r_{\mathrm{rsp}}}](R', S')$. Let $N \le m \le F$ with $2m > F$, and set*

$$A_m = [2^{F-m}]R, \quad B_m = [2^{F-m}]S, \quad C_m = [2^{F-m}]R', \quad D_m = [2^{F-m}]S'.$$

Let $\tau = t_{2^m}(A_m, B_m)$ and $\tau' = t_{2^m}(C_m, D_m)$. Then τ has order 2^{2m-F}, $\tau'^{-1} \in \langle \tau \rangle$, and $y_m = \log_\tau(\tau'^{-1})$ satisfies

$$\det \mathbf{M} \;\equiv\; 2^{r_{\mathrm{rsp}}}\, y_m + 2^{g-1} \pmod{2^{\min(g,\, 2m-F+r_{\mathrm{rsp}})}}.$$

Proof. Let $\varepsilon = e_{2^F}(R, S)$ and $\varepsilon' = e_{2^F}(R', S')$. Both are primitive 2^F-th roots of unity, so $\varepsilon'^{-1} = \varepsilon^\lambda$ for a unique $\lambda \in \mathbb{Z}/2^F\mathbb{Z}$. *Weil side.* By (W4) and (W1), the pairings of Theorem 1 are $\omega = \varepsilon^{2^{F-N}}$ and $\eta = \varepsilon'^{2^{F-N+r_{\mathrm{rsp}}}}$, so Step 4 of its proof reads

$$\zeta \;=\; \omega^{\det \mathbf{M}}\, \eta \;=\; \varepsilon^{\,2^{F-N}(\det \mathbf{M} - 2^{r_{\mathrm{rsp}}}\lambda)},$$

a primitive fourth root of unity. Hence $\det \mathbf{M} - 2^{r_{\mathrm{rsp}}}\lambda \equiv 2^{N-2}u \pmod{2^N}$ with u odd, and since $N - 1 = g$,

$$\det \mathbf{M} \;\equiv\; 2^{r_{\mathrm{rsp}}}\lambda + 2^{g-1} \pmod{2^g}. \tag{4}$$

Tate side. By (W4), $e_{2^m}(A_m, B_m) = e_{2^F}\big(R, [2^{F-m}]S\big) = \varepsilon^{2^{F-m}}$. Likewise $e_{2^m}(C_m, D_m) = \varepsilon'^{2^{F-m}}$. By Lemma 2, applied to E_{chl} and to E_{aux} with the same exponent,

$$\tau = \varepsilon^{-c\,2^{2F-2m}}, \qquad \tau' = \varepsilon'^{-c\,2^{2F-2m}}.$$

Since c is odd, τ has order $2^{F-(2F-2m)} = 2^{2m-F}$, and

$$\tau'^{-1} = \varepsilon'^{\,c\,2^{2F-2m}} = \varepsilon^{-\lambda c\,2^{2F-2m}} = \tau^\lambda \in \langle \tau \rangle.$$

Thus $y_m \equiv \lambda \pmod{2^{2m-F}}$, so $2^{r_{\mathrm{rsp}}} y_m \equiv 2^{r_{\mathrm{rsp}}}\lambda \pmod{2^{2m-F+r_{\mathrm{rsp}}}}$, and combining with (4) gives the claim. □

Corollary 2. *One Tate ladder of order 2^m per curve determines* $\det \mathbf{M} \bmod 2^g$ *whenever*

$$m \;\ge\; \left\lceil \frac{F + g - r_{\mathrm{rsp}}}{2} \right\rceil,$$

that is, with $r_{\mathrm{rsp}} = 0$ and $g = e_{\mathrm{rsp}} + 1$ in round 3, $m \ge 261, 402, 533$ at levels I, III, V; and $\det \mathbf{M} \bmod 2^{e_{\mathrm{rsp}}}$ *from $m \ge \lceil (F + e_{\mathrm{rsp}})/2 \rceil = 260, 401, 532$.*

Lemma 2 relates the two pairings exactly, with no square root, so no bit is lost and the threshold is a theorem. Table 3 gives, per level, the smallest m at which the ladder of Proposition 2 determines $\det \mathbf{M}$ modulo 2^g on all 100 KAT vectors, and modulo $2^{e_{\mathrm{rsp}}}$; both equal the corollary's bounds, one apart, and on every vector individually: at the bound less one the ladder determines $2m - F$ bits of λ, one short of the modulus, on every vector. At level I the verifier's pairing stage, the two bases from the hints at order 2^N (126 doublings per point after the cofactor) and two Weil pairings of two ladders of $N - 1$ steps each, takes 783.7 μs; the Tate alternative, the bases at order 2^{261} (63 doublings per point), one ladder of 260 steps per curve, and the 63 further doublings per point with a difference point that give the chain its order-2^N bases, takes 648.6 μs, 17% less. The discrete logarithm is the same size on both paths, $2m - F = N$ bits at the bound.

Table 3: Smallest ladder order 2^m at which one Tate ladder per curve on the scaled full-order bases determines $\det \mathbf{M}$ on all 100 KAT vectors of a level, modulo 2^g and modulo $2^{e_{\mathrm{rsp}}}$, against the bounds of Corollary 2.

Level	F	N	g	m, mod 2^g	$\lceil (F+g)/2 \rceil$	m, mod $2^{e_{\mathrm{rsp}}}$	$\lceil (F+e_{\mathrm{rsp}})/2 \rceil$
I	324	198	197	261	261	260	260
III	500	304	303	402	402	401	401
V	664	402	401	533	533	532	532

7 Costs

Table 4 is level I verification with `sqisign-rs` 0.7 on the 100 KAT vectors: for each vector, every component of each path timed on a copy of the state that path feeds it, the median of 200 iterations after 5 warm-ups, and then the median over the vectors; both paths under a prepared public key (the per-key basis work done once), on an Intel Xeon at 2.80 GHz (Cascade Lake, KVM guest) with `rustc` 1.99.0 and the crate's release profile (fat LTO, one codegen unit, no `target-cpu` flag; the x86-64 assembly field kernels selected at run time). Three runs agreed on every component to within 0.6% and on the full verifications to within 0.1%.

Table 4: Level I verification cost in microseconds under a prepared key, the median over the 100 KAT vectors of the per-vector medians: the standard path (**S**) on the decoded signature and the integrated compressed path (**C**) on the decoded compressed signature, each path's components timed on its own state.

Component	**S**	**C**
decoding	0.3	2.4
E_{aux} from A_{aux}	0.1	0.1
challenge derivation $E_{\mathrm{pk}} \to E_{\mathrm{chl}}$	583.5	583.4
bases from the hints at order 2^N, both curves	215.4	215.4
two Weil pairings		559.9
two inversions and the discrete logarithm		189.5
$\det \mathbf{M}$ and the dropped entry		0.1
matrix range and normal-form check	0.0	0.0
basis change, apply $\mathbf{M}$	368.0	365.9
$(2^{e_{\mathrm{rsp}}}, 2^{e_{\mathrm{rsp}}})$-chain to E_{com}	2908.5	2908.6
challenge hash and compare	15.3	15.3
sum of component medians	4091.2	4840.7
full verification from bytes, median	4103.3	4855.5

The integrated path, which the implementation runs, derives the challenge curve and

the two bases once; what it adds is the two pairings, the inversions and the discrete logarithm, the recovery and the larger decoding: 752 μs, 18% of the standard verification (4103 μs). With the Tate ladder of §6 in place of the two Weil ladders, the pairing stage measured there is 135 μs cheaper on entry 0, and the overhead falls to 617 μs, 15%.

8 What was checked

Table 5 lists the checks, all on the official NIST vectors of the reference implementation at tag `nist-v3` [SQI26], 100 per level, with `sqisign-rs` version 0.7 on branch `fix/compressed-nbt`. The tests and logs are in the repository [Ray26c] under `paper-weil/measurements/round3` and `crates/kat/tests/kat_compressed.rs`.

Table 5: Checks on the 300 round-3 KAT vectors (counts per level I / III / V).

Check	Result
Theorem 1: $y + 2^{g-1}$ and $\det \mathbf{M}$ agree modulo 2^g	100 / 100 / 100
they first differ at bit g ($u = 3$); they agree modulo 2^N ($u = 1$)	46 / 59 / 54; 54 / 41 / 46
$\omega^{\det \mathbf{M}}\eta$ is a primitive fourth root of unity	100 / 100 / 100
$\det \mathbf{M}$ odd; M_{00}, M_{01} both even	100 / 100 / 100; 0 / 0 / 0
M_{00} even (pivot path, M_{11} transmitted)	32 / 22 / 37
recomputed hints equal the transmitted ones	100 / 100 / 100
$[2^{F-N}]$ of the full-order basis of Proposition 2 is the verifier's basis	100 / 100 / 100
compressed round trip, verify, decompress and verify	100 / 100 / 100
decompressed matrix exact; else differs in bit $e_{\mathrm{rsp}} + 1$ only	14 / 14 / 13; 86 / 86 / 87
Tate on the verifier's bases determines $\det \mathbf{M}$ modulo $2^{e_{\mathrm{rsp}}}$, any m	0 / 0 / 0
one Tate ladder on the full-order bases, $m = \lceil (F + g)/2 \rceil$; m one less	100 / 100 / 100; 0 / 0 / 0

References

[AAA+25] M. A. Aardal, G. Adj, D. F. Aranha, A. Basso, et al. SQIsign: Algorithm specifications and supporting documentation, version 2.0. NIST PQC Round 2 submission, February 2025. URL: https://sqisign.org/spec/sqisign-20250205.pdf.

[AAA+26] M. A. Aardal, G. Adj, D. F. Aranha, A. Basso, et al. SQIsign: Algorithm specifications and supporting documentation, version 3.0. NIST PQC Round 3 submission, September 2026. URL: https://sqisign.org/spec/sqisign-20260901.pdf.

[ABF+25] M. A. Aardal, A. Basso, L. De Feo, S. Patranabis, and B. Wesolowski. A complete security proof of SQIsign. Cryptology ePrint Archive, Paper 2025/379, 2025. URL: https://eprint.iacr.org/2025/379.

[BDF+24] A. Basso, P. Dartois, L. De Feo, A. Leroux, L. Maino, G. Pope, D. Robert, and B. Wesolowski. SQIsign2D-West: The fast, the small, and the safer. In *ASIACRYPT 2024, Part III*, volume 15486 of *Lecture Notes in Computer Science*, pages 339–370. Springer, 2024.

[BMP23] A. Basso, L. Maino, and G. Pope. FESTA: Fast encryption from supersingular torsion attacks. In *ASIACRYPT 2023, Part VII*, volume 14444 of *Lecture Notes in Computer Science*, pages 98–126. Springer, 2023.

[CJL+17] C. Costello, D. Jao, P. Longa, M. Naehrig, J. Renes, and D. Urbanik. Efficient compression of SIDH public keys. In *EUROCRYPT 2017, Part I*, volume 10210 of *Lecture Notes in Computer Science*, pages 679–706. Springer, 2017.

[DLRW24] P. Dartois, A. Leroux, D. Robert, and B. Wesolowski. SQIsignHD: New dimensions in cryptography. In *EUROCRYPT 2024, Part I*, volume 14651 of *Lecture Notes in Computer Science*, pages 3–32. Springer, 2024.

[Kan97] E. Kani. The number of curves of genus two with elliptic differentials. *J. für die reine und angewandte Mathematik*, 485:93–122, 1997.

[NO24] K. Nakagawa and H. Onuki. QFESTA: Efficient algorithms and parameters for FESTA using quaternion algebras. In *CRYPTO 2024, Part V*, volume 14924 of *Lecture Notes in Computer Science*, pages 75–106. Springer, 2024.

[PB21] G. C. C. F. Pereira and P. S. L. M. Barreto. Isogeny-based key compression without pairings. In *PKC 2021*, volume 12710 of *Lecture Notes in Computer Science*, pages 131–154. Springer, 2021.

[Ray26a] D. Ray. Auxiliary isogeny freedom in SQIsign's two-dimensional representation. Cryptology ePrint Archive, Paper 2026/1305, 2026. URL: https://eprint.iacr.org/2026/1305.

[Ray26b] D. Ray. Technical notes on signature compression techniques. Issue 19 of `SQISign/the-sqisign`, 2026. Opened July 8, 2026, with replies by L. De Feo of July 8 and October 7, 2026. URL: https://github.com/SQISign/the-sqisign/issues/19.

[Ray26c] D. Ray. `sqisign-rs`: SQIsign in pure Rust. Version 0.7 (round 3), branch `fix/compressed-nbt`, 2026. URL: https://github.com/anchorageoss/sqisign-rs.

[Sch05] E. F. Schaefer. A new proof for the non-degeneracy of the Frey–Rück pairing and a connection to isogenies over the base field. In T. Shaska, editor, *Computational Aspects of Algebraic Curves*, volume 13 of *Lecture Notes Series on Computing*, pages 1–12. World Scientific, 2005.

[Sil86] J. H. Silverman. *The Arithmetic of Elliptic Curves*, volume 106 of *Graduate Texts in Mathematics*. Springer, 1986.

[SQI26] SQIsign team, The. `the-sqisign`, reference implementation, tag `nist-v3`. Commit `6d01770`, September 2026. URL: https://github.com/SQISign/the-sqisign.

[Wes26] B. Wesolowski. The supersingular isogeny problem in time and memory $p^{1/3+o(1)}$. Cryptology ePrint Archive, Paper 2026/1486, 2026. URL: https://eprint.iacr.org/2026/1486.

[ZSP+18] G. H. M. Zanon, M. A. Simplicio Jr., G. C. C. F. Pereira, J. Doliskani, and P. S. L. M. Barreto. Faster isogeny-based compressed key agreement. In *PQCrypto 2018*, volume 10786 of *Lecture Notes in Computer Science*, pages 248–268. Springer, 2018.